Lecture plan · for the lecturer

Learn from reward

University, scale 1

Time
45 min

Materials
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

Learning goals

- The reward is at the end
- Early steps must learn too

Activity: In the lab

1. Point at one shading (value) and its arrow (policy)
2. Work one update on paper

Discussion

- Can a reward be wrong?

Exit ticket

- Value is not policy. Finish the sentence.

Try it in your next lecture
Bring that wrong reward to the seminar before the lab.

Seminar sheet · for students

Learn from reward

University, scale 1

Name _____ Group _____ Date _____

1 **A reward arrives after a path of actions. The hard part is…**

- ◯ Sending the credit back along the path
- ◯ Giving every step the same reward
- ◯ Making the final reward bigger

2 **A value and a policy are the same picture.**

- ◯ True
- ◯ False

3 **Match the word.**

Reward	• • • •	How good a place is
Value	• • • •	What to do there
Policy	• • • •	The number the world returns

4 **Order one learning step from reward.**

- ☐ See the new place and the reward
- ☐ Update the value of where you were
- ☐ Take an action

Answers · for the lecturer

Learn from reward

University, scale 1

1 **Sending the credit back along the path**

The reward comes only at the charger. A bigger reward, or the same one for every step, does not tell the early moves that they helped: the credit has to travel back along the path.

2 **False**

Value is how good it is to be here. Policy is what to do here.

3 **Reward — The number the world returns**
Value — How good a place is
Policy — What to do there

A reward that looks sensible can still train a bad policy.

4 **Top to bottom: 2, 3, 1**

The update is last. It needs the reward.

For you to think about

Which pair of words will you not let students swap?